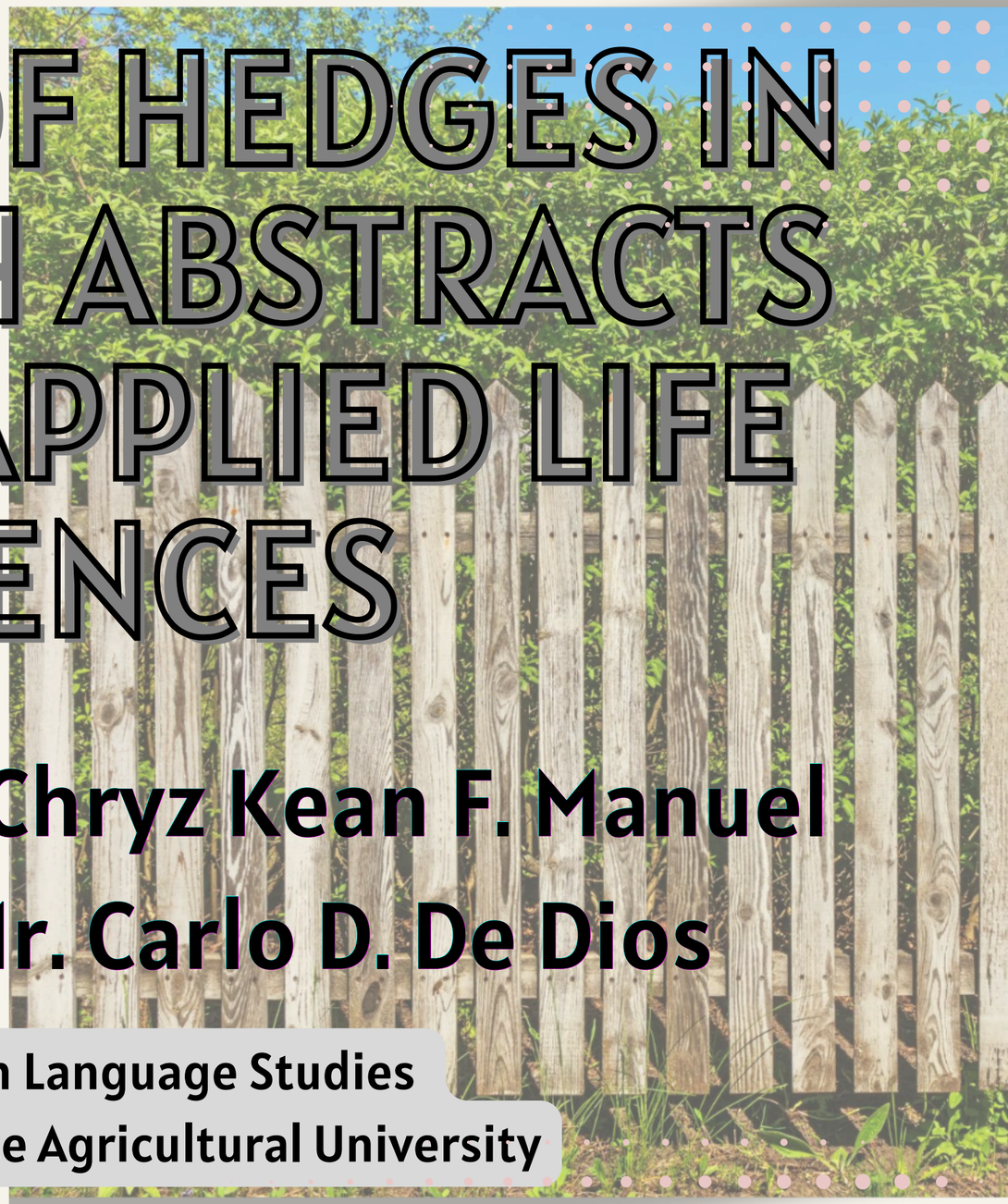
THE USE OF HEDGES IN RESEARCH ABSTRACTS ACROSS APPLIED LIFE SCIENCES



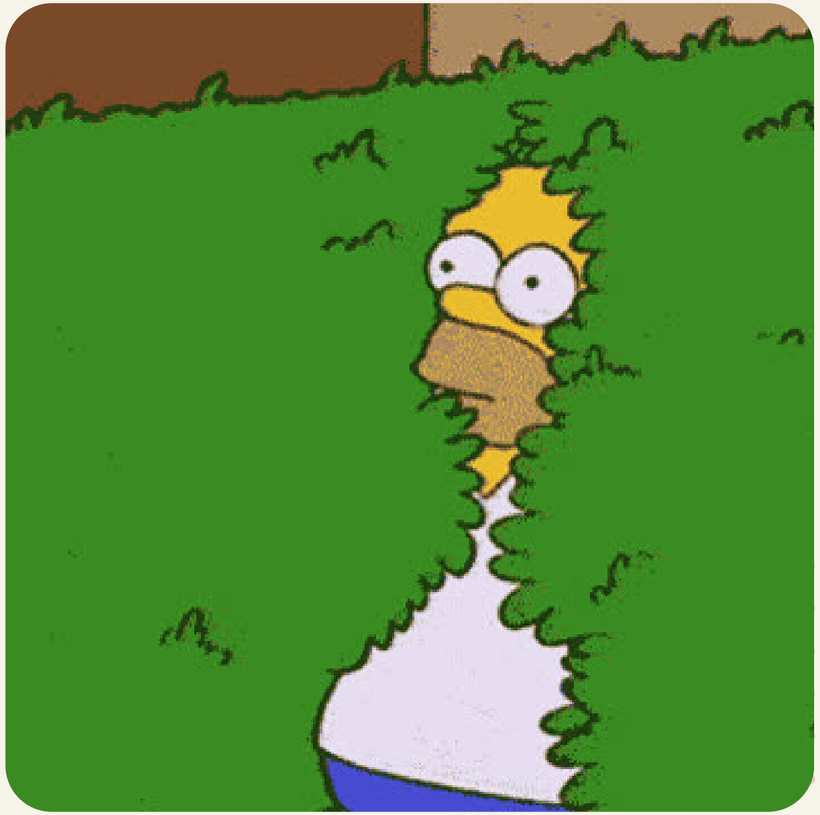
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INTRODUCTION

Overview

- Researchers are told to “qualify your claims,” but hedging shifts with evidence and field

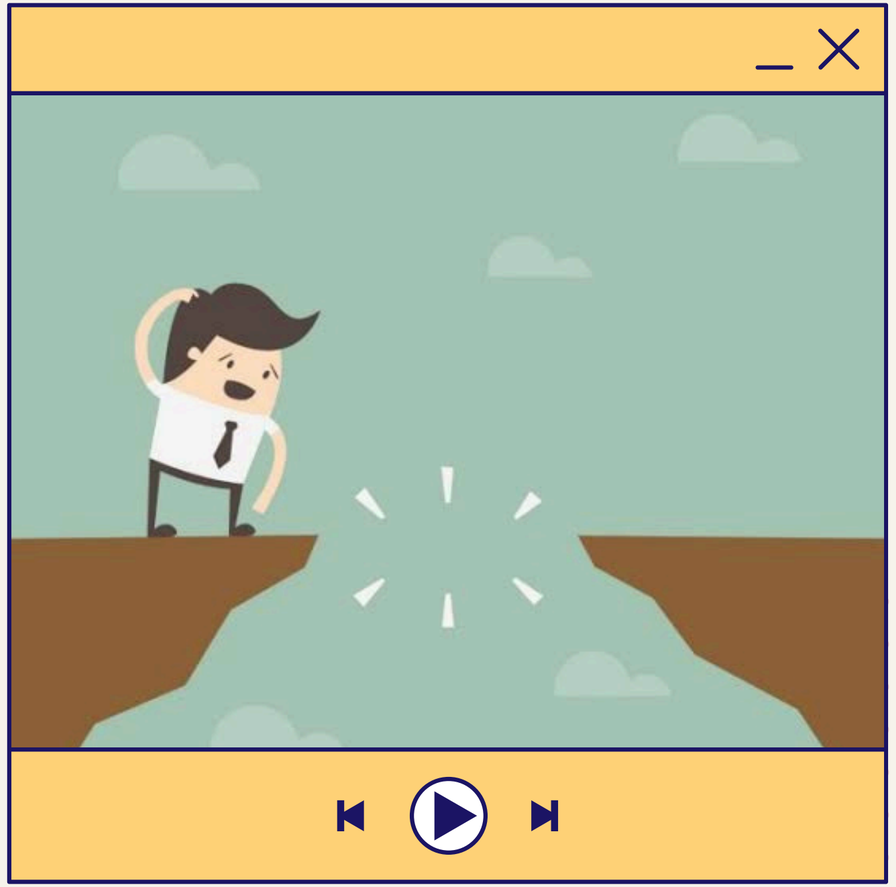


INTRODUCTION

- An ELS study of how stance is worded in L2 academic English
- Hedging is a core rhetorical resource in science writing (Hyland, 1996, 1998)
- HEDGE, as defined here: is a device that changes the writer's stance toward a proposition
- Undergraduate thesis writers face the same genre demands with less training, and little is known about them

Research Gap

- Most student hedging studies focus on English or language majors
- No published study covers applied science thesis abstracts in a Philippine L2 context



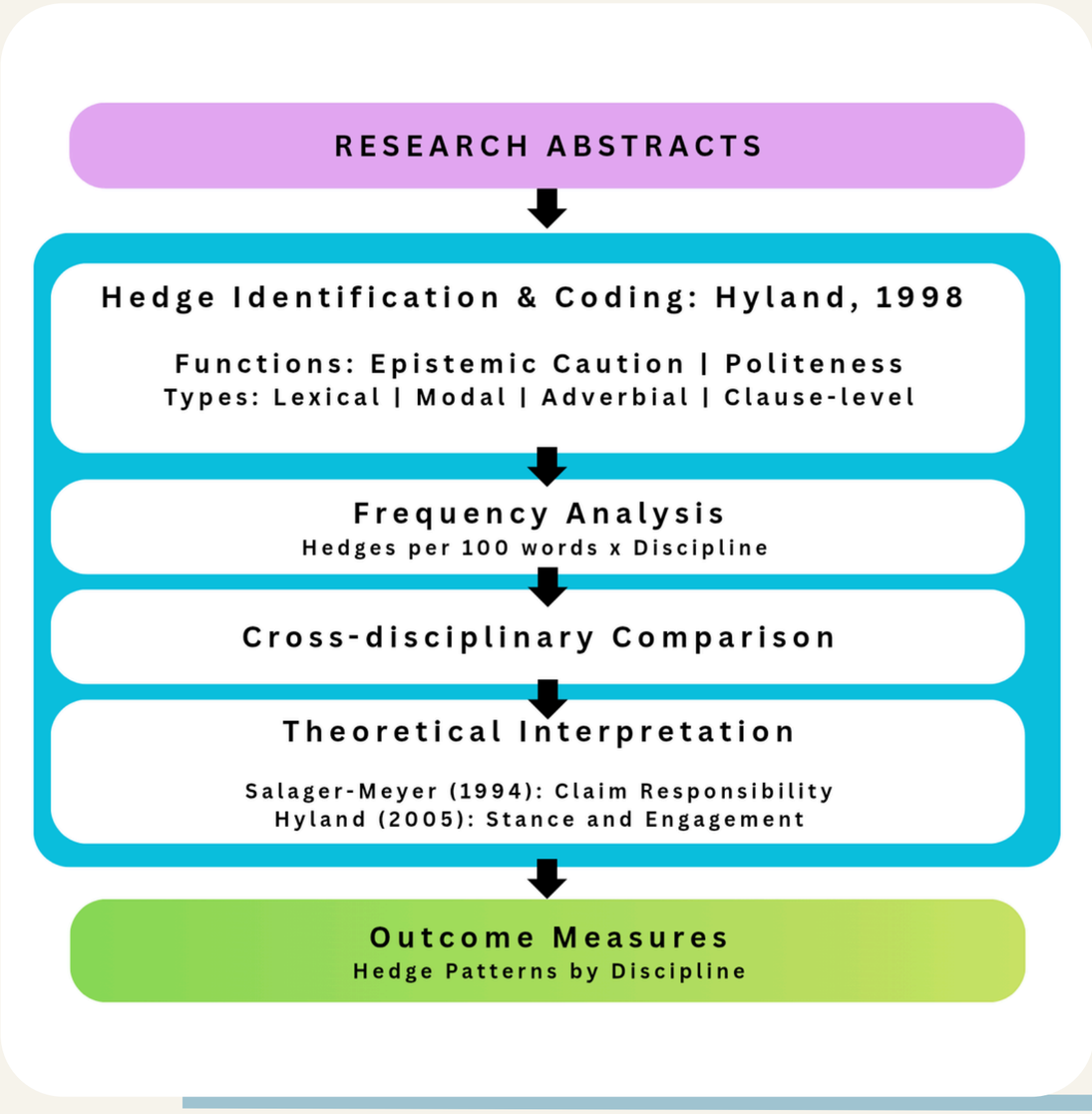


Research Questions

- What types of hedging devices appear across the three disciplines?
- How does hedge frequency vary across the programs?
- What rhetorical functions do the hedges serve?
- What do these patterns suggest about epistemic stance in the abstracts?

Framework

- Coding scheme: Hyland (1998), four hedge types (lexical, modal, adverbial, clause-level)
- Two functions: Epistemic Caution (EC) and Politeness/Deference (P)
- Interpretive lenses: Salager-Meyer (1994) claim responsibility and Hyland (2005) stance model



METHODOLOGY



Design

- Descriptive mixed methods: frequency counts plus functional interpretation
- Hedges per 100 words for comparison across abstracts

Corpus

- 60 abstracts: Biology 20, Vet Med 20, Agriculture 20 (10 Animal Science, 10 Crop Science)
- Disproportionate stratified systematic sampling from a 381-thesis pool
- English-language only, hard-bound submission year 2023-2025



Process

- Permission, on-site photography, OCR, manual verification
- Pilot on 4 abstracts outside the corpus, then full coding
- Three-step coding: candidates, claim condition plus removal tests, type assignment



CODING EXAMPLE

Applying Step 2

Example:

As prior studies **indicated**, results **suggest** that fermented banana peel **can** significantly enhance growth.



Token I: indicated

Host clause: As prior studies ***indicated***, results suggest that fermented banana peel can significantly enhance growth.

(The attribution frames the whole finding against prior work.)

I. Claim condition: Is this about a knowledge claim on findings?

– Yes. It reports what the results support, in relation to prior studies.

Removal checks (remove only suggest)

Version	Text
With token	As prior studies **indicated**, results suggest that fermented banana peel can significantly enhance growth.
Without token	Results suggest that fermented banana peel can significantly enhance growth.

Working Test. Does commitment to the finding changed?

- **No**. Truth commitment is unchanged.

Reader Test. What changes for the reader?

- The writer **no longer shares** the finding with prior studies.
- The writer now **owns** the claim alone.

So... Working Test fails, Reader Test passes. Retained hedge.

Type = Lexical. Step 3 = P.

Token 2: suggest

Host clause: Results ***suggest*** that fermented banana peel can significantly enhance growth.

(The softening sits on how the finding is introduced.)

1. Claim condition: Is this about a knowledge claim on findings?

– Yes, it reports what the results support.

Removal checks (remove only suggest)

Version	Text
With token	Results **suggest** that fermented banana peel can significantly enhance growth.
Without token	Results [∅] that fermented banana peel can significantly enhance growth. → repaired as *Results show that… / Results are that… /* bare *Fermented banana peel can…*

What changes?

- Commitment to the finding **gets stronger** (less distance between writer and claim).
- The claim is **more direct**.

So...

Pass. Retained hedge. Type = Lexical. Step 3 = EC.

Token 3: can

Host clause: fermented banana peel ***can*** significantly enhance growth.

(inside the that-clause)

I. Claim condition

Still a findings/implication claim? Yes.

Removal checks (remove only suggest)

Version	Text
With token	… peel **can** significantly enhance growth.
Without token	… peel significantly enhance(s) growth.

What kind of change is this?

In codebook, can + effect verb in this pattern is usually capability (what the treatment is able to do), not "maybe it does."

- **Working Test: removing can does not mainly raise epistemic caution about whether the result is true. It reads as ability/potential of the treatment.**
- **Reader Test: no prior work is attributed, and no softening toward the reader is lost.**

So... Fails both tests. Code FP-Cap.

RESULTS, ANALYSIS, & DISCUSSION

RQ1: Types (see Tables 2 & 4)

- 65 hedges total, about 1.08 per abstract
- Lexical 64.6% (42), modal 30.8% (20), adverbial 4.6% (3), clause-level 0
- “Potential” alone is 19 of 65 hedges.
- Differs from the modal-heavy pattern in the literature
- Without “potential,” lexical (23) and modal (20) are nearly even

RQ2: Frequency (see Table 5)

- Agriculture 0.63, Biology 0.62, Vet Med 0.20 hedges per 100 words
- 25 of 60 abstracts (41.7%) had zero hedges; 12 of them were Vet Med
- Agriculture's lead comes from one outlier abstract (Acs2, 8 hedges)
- Low density is common in L2 undergraduate writing (Hyland & Milton, 1997; Qiu et al., 2024)

RQ3: Functions (see Table 6)

- All 65 retained hedges were Epistemic Caution; Politeness = 0
- 65 false positives were excluded
- Most retained hedges sit in implication and conclusion sentences. Results sentences often use factive verbs (showed, revealed, demonstrated).
- An all-EC result fits how scientific English uses hedges on knowledge claims

- Abstracts leave less room for interpersonal markers (Gillaerts & Van de Velde, 2010)
- Hyland (1996, 1998): scientific hedging is a resource for precision, caution, and negotiated commitment to propositions
- Salager-Meyer (1994): hedges distribute responsibility for a statement's accuracy or reach among the writer, the data, and external conditions
- The result is a functionally narrow stance repertoire centered on the proposition

RQ4: Epistemic stance (Table 7)

- Lenses applied: Salager-Meyer (1994) claim responsibility and Hyland (2005) stance model
- Binary outcomes do not explain Vet Med's low rate: Biology binary abstracts average 1.57 hedges vs. 0.38 in Vet Med
- Salager-Meyer: fewer hedges on implications means the writer takes fuller responsibility for those claims
- Hyland (2005): fewer hedges means less explicit softening of authorial commitment

- Vet Med shows no extra overclaiming (4 unhedged assertions, same as Biology), so the difference is in how caution is worded
- Acs2 shows rescue-hedging: responsibility is redistributed to keep a non-significant treatment sounding viable



CONCLUSIONS & RECOMMENDATIONS

Conclusions

- Hedging is constrained: results are often stated plainly, and hedges concentrate on implications
- The repertoire is narrow and lexical, led by “potential” and “suggest”
- Disciplinary difference exists but is thin; Vet Med is the clear low-hedge group
- Texts are adviser-ratified, so the patterns partly reflect program practice

- For ELS: a baseline for how L2 undergraduates in applied sciences word stance in abstracts
- For BAELS and related instruction: shows what hedging in science abstracts looks like on the ground at this campus (uneven & lexically narrow)
- For research: makes later comparison possible against expert abstracts, other programs, and fuller thesis chapters, where other functions of mitigation may have more room

Recommendations

- Advisers and instructors: focus feedback on the move from results to implications. Teach students to match claim strength to evidence strength.
- Future researchers: proficiency measures, comparison to expert abstract corpus
- Apply the coding scheme to longer genres, such as discussion chapters

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THANK YOU